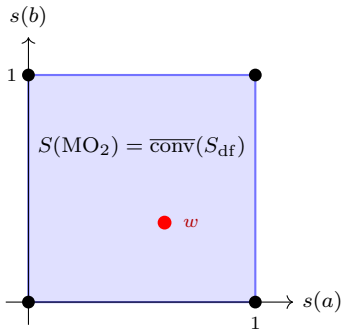


(1) MO_2

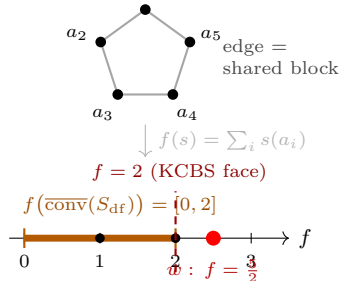
corners fill the state space



hull of the 4 corners = the whole square;
every state is a mixture of corners
 $\Rightarrow$ **no contextual state**

(2) Pentagon

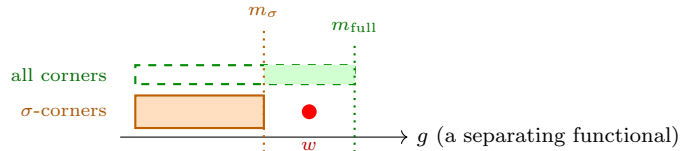
the lattice, and its hull projected



every dispersion-free corner has $f = \sum s(a_i) \leq 2$
(odd loop: no two adjacent atoms both 1), so every
mixture does; but $w \equiv \frac{1}{2}$ has $f = \frac{5}{2} \Rightarrow$ **contextual**

(3) σ -essential

the leak between two hulls



the σ -additive corners reach only $g \leq m_\sigma$; the
extra “fake” corners push the full hull to $g \leq m_{\text{full}}$.
 w has $m_\sigma < g(w) \leq m_{\text{full}}$: **outside** $\overline{\text{conv}}(S_{\text{df}}^\sigma)$,
inside $\overline{\text{conv}}(S_{\text{df}})$ — and no finite face sees it